Grading Criteria: There are eight test cases, taken from the ISPD 2010 High Performance Clock Network Synthesis Contest and simplified.

編號	ISPD	sink 數	晶片	s	SRC	SRC length	w_skew	buffer library (F/L/C)
001	01	1,107	8000×8000	240	(400, 400)	16000	5	4/1443/5, 10/4328/14, 28/10099/40
002	02	2,249	13000×7000	201	(650, 350)	20000	25	4/1207/5, 10/3621/14, 28/8448/40
003	03	1,200	3072×493	36	(154, 25)	3565	100	4/213/5, 10/639/14, 28/1492/40
004	04	1,845	2130×2690	56	(106, 134)	4820	5	4/334/5, 10/1003/14, 28/2341/40
005	05	1,016	2319×2545	76	(116, 127)	4864	25	4/457/5, 10/1372/14, 28/3201/40
006	06	981	1950×891	42	(98, 45)	2841	100	4/253/5, 10/758/14, 28/1768/40
007	07	1,915	2537×1448	44	(127, 72)	3985	5	4/263/5, 10/788/14, 28/1840/40
008	08	1,134	1837×1628	51	(92, 81)	3465	25	4/308/5, 10/924/14, 28/2157/40

In general, complexity scales with the number of sinks. Some students' programs fail to finish on these cases and simply crash, which would normally mean a score of 0. However, since too many students would end up with 0 that way, I will also run the input1 test case from project2.pdf. If your program produces a result on it, you receive a guaranteed minimum of 40 points. The intent is to give you a chance to catch up in later projects (Project 3, Project 4, Project 5, etc.) instead of failing the course.

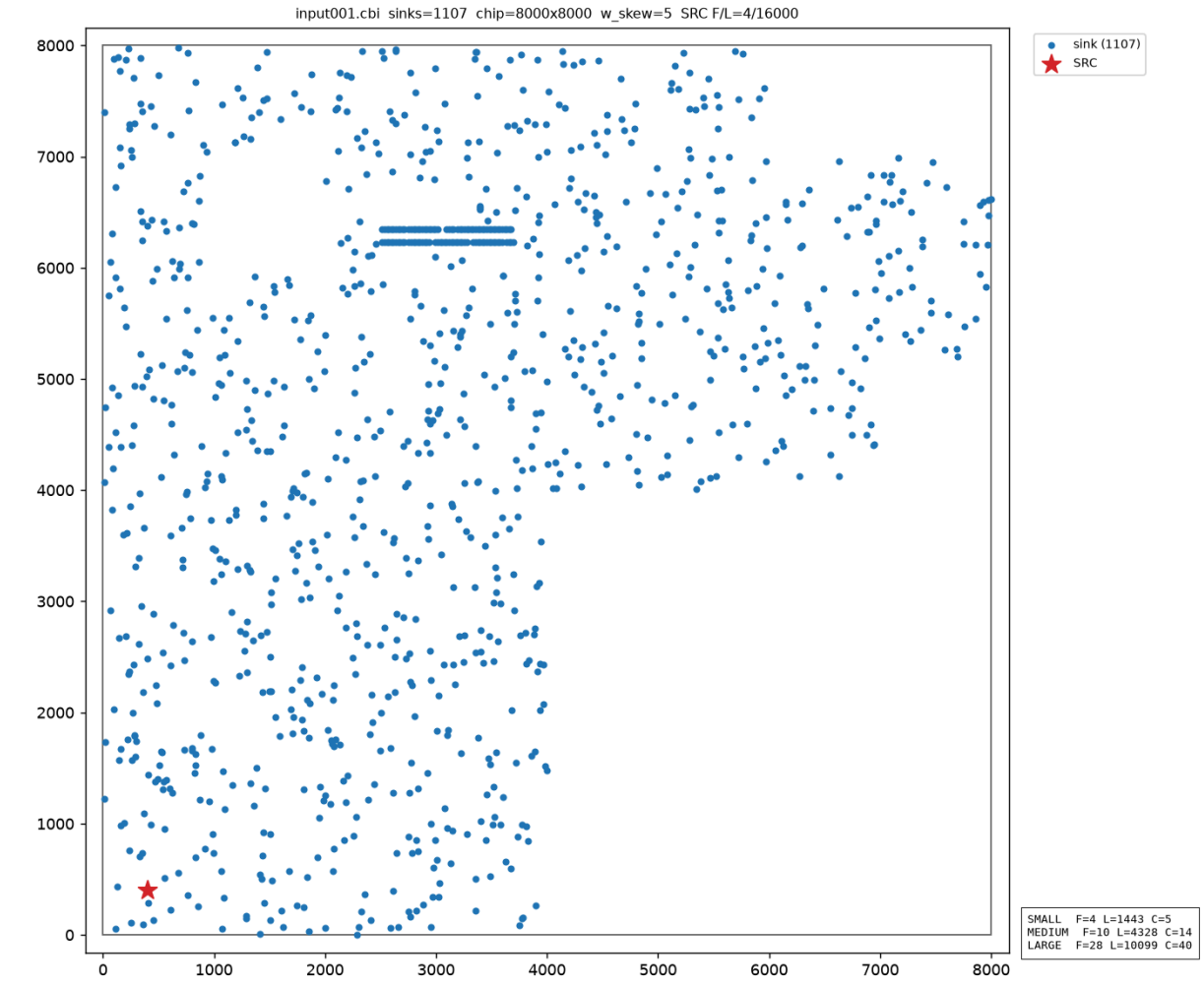
Overall, everyone will be divided into three groups:

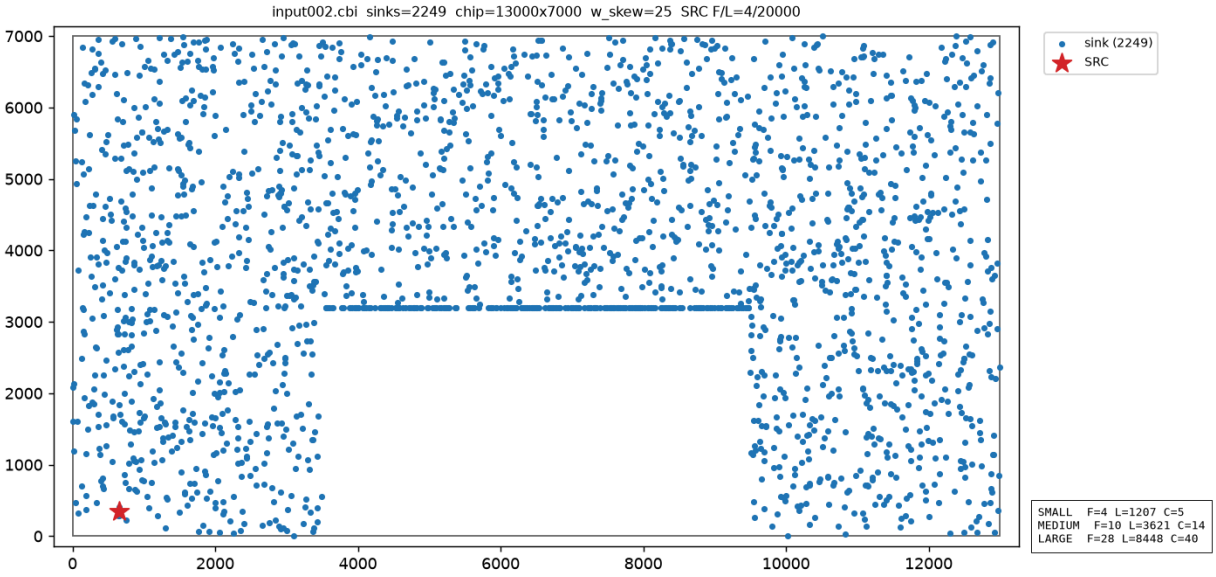
1. Group A: Passes at least six of the eight test cases. Final scores are interpolated within this group, with a minimum of 70.
2. Group B: Passes zero to five of the eight test cases, but the output from project2.pdf passes the verifier. The base score is 40, and each additional passed test case adds 4 points, for a minimum of 40 and a maximum of 60.
3. Group C: The output from project2.pdf does not pass the verifier.

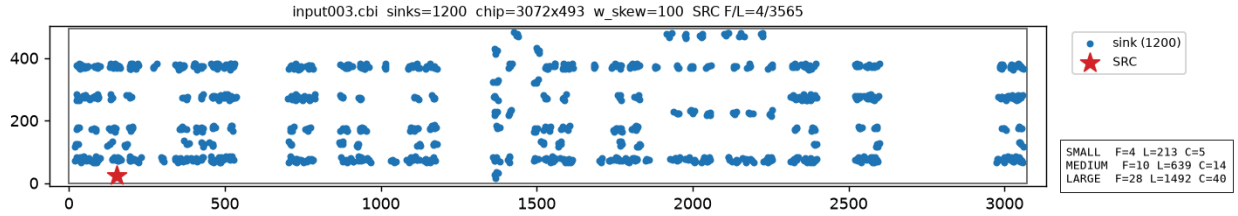
The scores above are program scores. Multiply by 0.9, then add the report score (0–10 points) to get the final grade on Moodle.

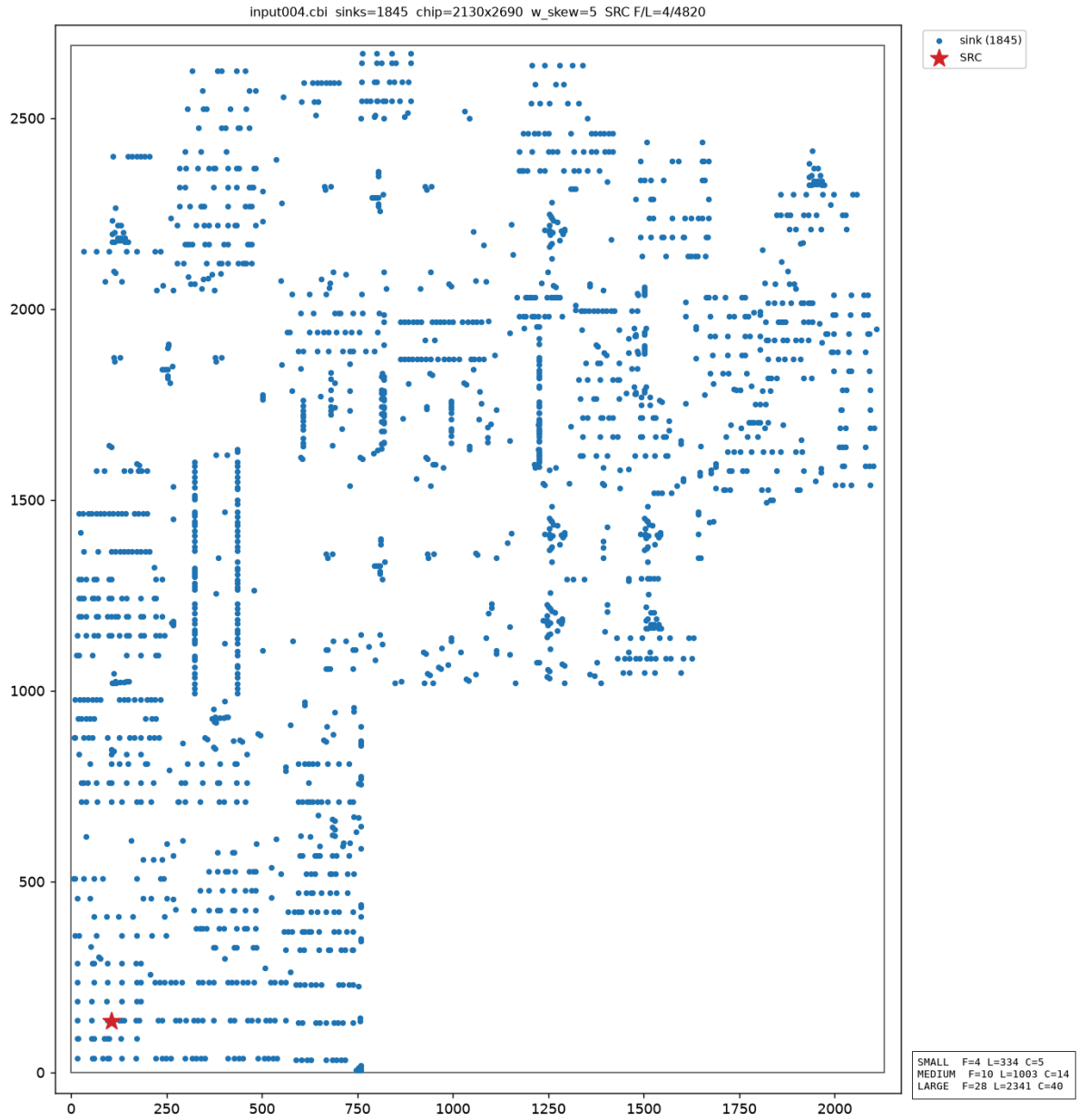
Some may think the TA should announce the test case sizes in advance, but you should be aware of your own algorithm's complexity.

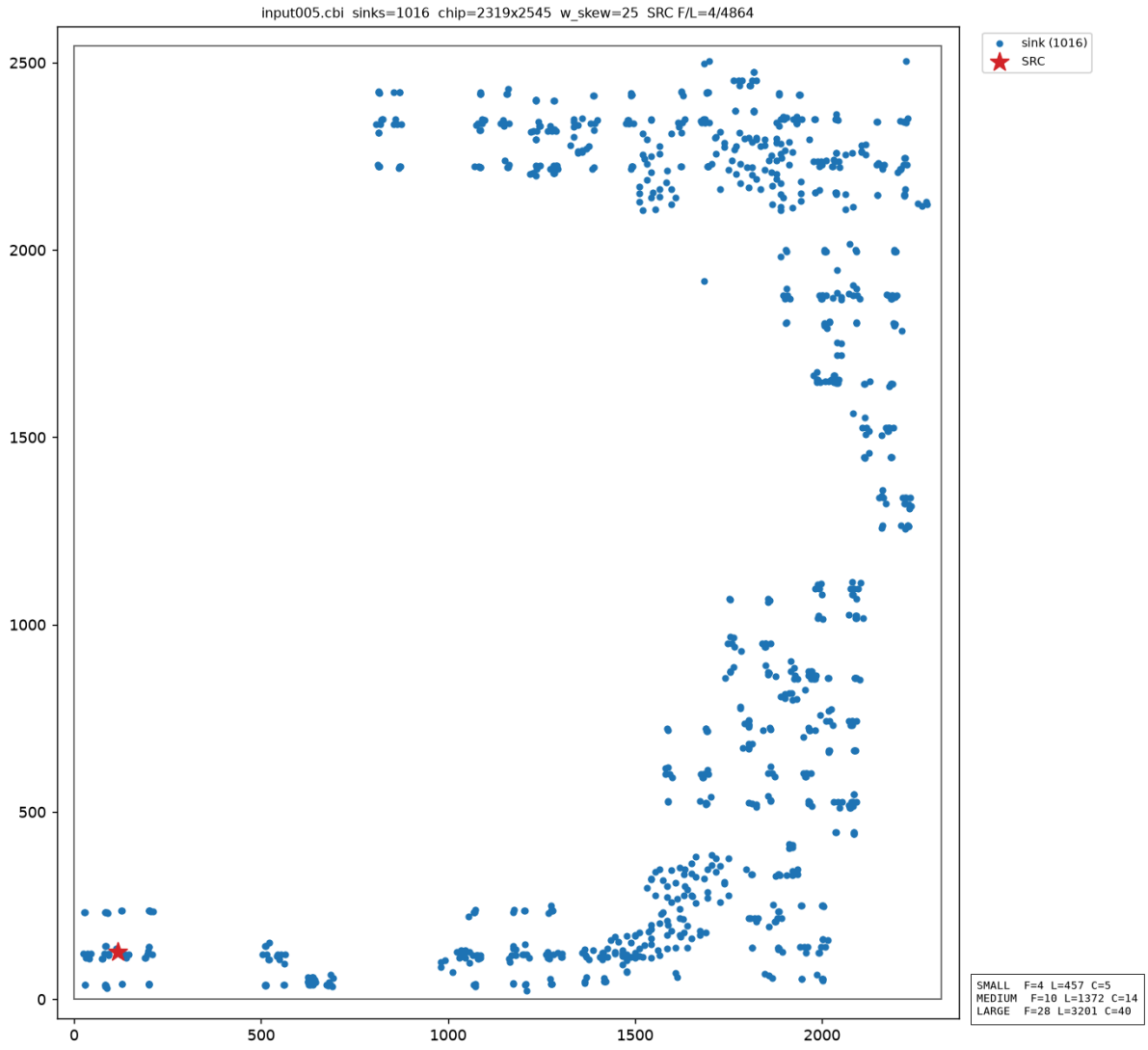
Test case images:

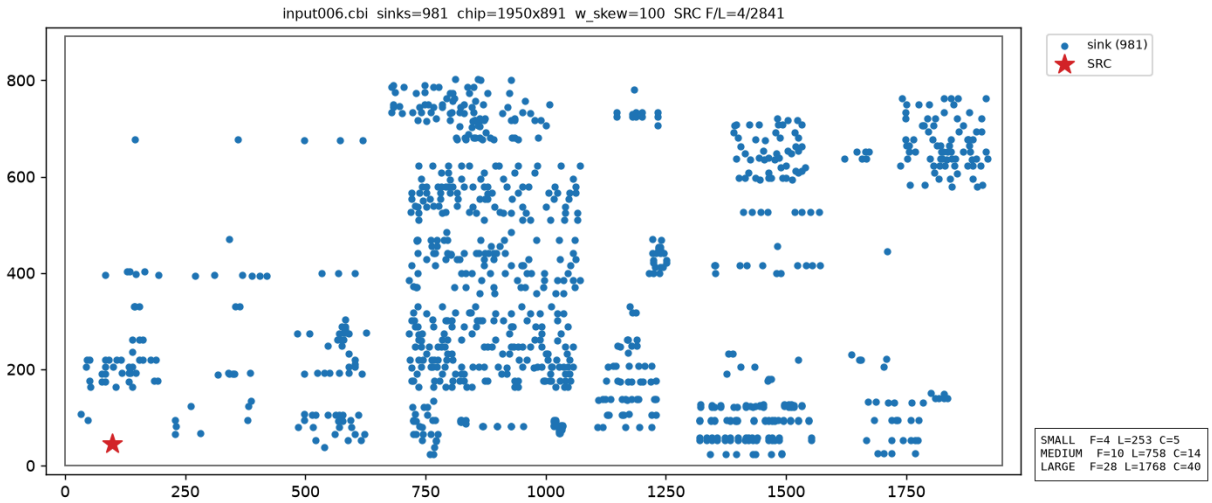

input001.cbi sinks=1107 chip=8000x8000 w_skew=5 SRC F/L=4/16000
sink (1107)
SRC
SMALL F=4 L=1443 C=5
MEDIUM F=10 L=4328 C=14
LARGE F=28 L=10099 C=40
0
1000
2000
3000
4000
5000
6000
7000
8000

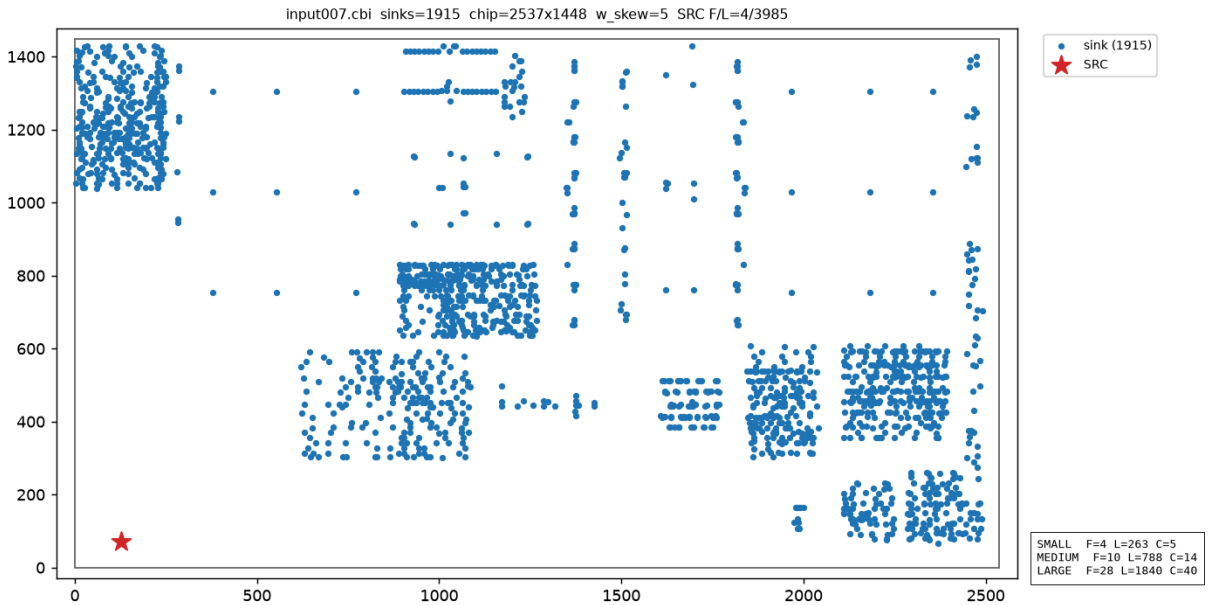

input002.cbi sinks=2249 chip=13000x7000 w_skew=25 SRC F/L=4/20000
sink (2249)
SRC
SMALL F=4 L=1207 C=5
MEDIUM F=10 L=3621 C=14
LARGE F=28 L=8448 C=40
0
1000
2000
3000
4000
5000
6000
7000
0
2000
4000
6000
8000
10000
12000


input003.cbi sinks=1200 chip=3072x493 w_skew=100 SRC F/L=4/3565
sink (1200)
SRC
SMALL F=4 L=213 C=5
MEDIUM F=10 L=639 C=14
LARGE F=28 L=1492 C=40
0
200
400
0
500
1000
1500
2000
2500
3000


input004.cbi sinks=1845 chip=2130x2690 w_skew=5 SRC F/L=4/4820
sink (1845)
SRC
SMALL F=4 L=334 C=5
MEDIUM F=10 L=1003 C=14
LARGE F=28 L=2341 C=40
0
500
1000
1500
2000
2500
0
250
500
750
1000
1250
1500
1750
2000


input005.cbi sinks=1016 chip=2319x2545 w_skew=25 SRC F/L=4/4864
sink (1016)
SRC
SMALL F=4 L=457 C=5
MEDIUM F=10 L=1372 C=14
LARGE F=28 L=3201 C=40
0
500
1000
1500
2000
2500


input006.cbi sinks=981 chip=1950x891 w_skew=100 SRC F/L=4/2841
sink (981)
SRC
SMALL F=4 L=253 C=5
MEDIUM F=10 L=758 C=14
LARGE F=28 L=1768 C=40
0
200
400
600
800
0
250
500
750
1000
1250
1500
1750


input007.cbi sinks=1915 chip=2537x1448 w_skew=5 SRC F/L=4/3985
sink (1915)
SRC
SMALL F=4 L=263 C=5
MEDIUM F=10 L=788 C=14
LARGE F=28 L=1840 C=40
0
200
400
600
800
1000
1200
1400
0
500
1000
1500
2000
2500


input008.cbi sinks=1134 chip=1837x1628 w_skew=25 SRC F/L=4/3465
sink (1134)
SRC
SMALL F=4 L=308 C=5
MEDIUM F=10 L=924 C=14
LARGE F=28 L=2157 C=40
0
200
400
600
800
1000
1200
1400
1600
0
250
500
750
1000
1250
1500
1750